



IILM UNIVERSITY, GREATER NOIDA

Department of Computer Science & Engineering

INTERNSHIP REPORT

on

“Development of MyDocs: A Secure Cloud-Based Document Management System”

Submitted in partial fulfilment of the requirements for the
Degree of Bachelor of Technology in
Computer Science & Engineering

Submitted by

Aryan Kumar

Roll No. 25SCS1003004027

B.Tech Computer Science & Engineering - Section 2CSE29

Internship Organization: Codec Technologies Pvt. Ltd.

18 June 2026 - 18 July 2026

DECLARATION

I, Aryan Kumar, Roll No. 25SCS1003004027, a student of B.Tech Computer Science & Engineering, Section 2CSE29, IILM UNIVERSITY, GREATER NOIDA, hereby declare that the internship report titled “Development of MyDocs: A Secure Cloud-Based Document Management System” is an authentic record of the work carried out by me during my one-month internship as a Web Developer Intern at Codec Technologies Pvt. Ltd., from 18 June 2026 to 18 July 2026.

The information, project work and results presented in this report are based on my genuine effort and practical exposure gained during the internship period. This report has been prepared solely for academic purposes and has not been submitted elsewhere for the award of any other degree or diploma.

Place: Greater Noida

Date: 18 July 2026

Aryan Kumar

Roll No. 25SCS1003004027

B.Tech CSE, Section 2CSE29

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to Codec Technologies Pvt. Ltd. for providing me the opportunity to undergo a one-month internship as a Web Developer Intern and for the practical exposure and guidance extended to me throughout the internship period, from 18 June 2026 to 18 July 2026.

I am thankful to the assigned project head(s) at Codec Technologies for their consistent support and valuable feedback, which helped me understand real-world full-stack development practices and successfully develop my project, MyDocs - a secure system for storing, sharing and collaborating on documents.

I would also like to express my heartfelt gratitude to the Department of Computer Science & Engineering, IILM UNIVERSITY, GREATER NOIDA, for providing the academic foundation and encouragement that made this internship possible. I sincerely thank the faculty members of the department for their continuous guidance and support throughout my B.Tech programme.

Finally, I extend my thanks to my family and friends for their constant encouragement, patience and motivation during the course of this internship and while preparing this report.

Aryan Kumar

INTERNSHIP COMPLETION CERTIFICATE



Internship Completion Certificate - Codec Technologies Pvt. Ltd.

INTERNSHIP OFFER LETTER



INTERNSHIP OFFER LETTER

Dear Aryan Kumar,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Web Developer Intern
- Location & Mode: Pan India, Hybrid
- Duration : 18/06/2026 to 18/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

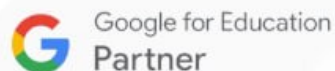
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 18/06/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



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Chandivali IT hub, Mumbai

Internship Offer Letter - Codec Technologies Pvt. Ltd.

ABSTRACT

This report presents an account of the one-month internship undertaken at Codec Technologies Pvt. Ltd. as a Web Developer Intern, from 18 June 2026 to 18 July 2026, in Hybrid mode under the Web Development / Full-Stack Web Development domain. The internship provided practical, industry-oriented exposure to modern full-stack web technologies, including Next.js, Node.js with Express, PostgreSQL and AWS S3, and complemented the theoretical knowledge acquired during the B.Tech Computer Science & Engineering programme at IILM University, Greater Noida.

As part of the internship, a project titled “Development of MyDocs: A Secure Cloud-Based Document Management System” was undertaken. MyDocs is designed as a centralized platform for storing, sharing and collaborating on documents. The system combines a Next.js interface, a Node.js/Express REST API, PostgreSQL database management through Prisma, and cloud file storage using AWS S3. Key capabilities include secure authentication, document upload and version control, role-based permissions, expiring share links, real-time collaboration, file encryption, audit logging, advanced search and tagging.

The report documents the weekly progression of work, the technical implementation of the MyDocs system across frontend and backend, the testing carried out to verify document management, security, sharing and collaboration behaviour, and the key technical and professional skills gained during the internship. The experience strengthened the author's practical understanding of full-stack development, relational database design, cloud storage integration and the development of secure, collaborative web applications.

Project focus

Centralized document storage + controlled sharing + version history + cloud storage + collaborative workflows + auditability.

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Report structure

The report follows the internship format used in the reference report, with preliminary pages, weekly work, project documentation, learning outcomes, challenges, future scope, conclusion and references.

CHAPTER 1: INTRODUCTION

1.1 Background of the Internship

Industrial internships form an essential component of a B.Tech Computer Science & Engineering programme, bridging the gap between classroom learning and real-world professional practice. While academic coursework builds a strong theoretical foundation in programming, databases and web technologies, an internship allows a student to apply that knowledge in a practical, outcome-driven setting. This experience helps students understand how individual concepts come together to produce a working software product.

As part of this academic requirement, I undertook a one-month internship as a Web Developer Intern at Codec Technologies Pvt. Ltd., from 18 June 2026 to 18 July 2026, in Hybrid mode. The internship focused on Full-Stack Web Development, giving me hands-on exposure to building a complete web application from the ground up using Next.js, Node.js with Express, PostgreSQL and AWS S3.

1.2 Objectives of the Internship

- To gain practical, hands-on experience in full-stack web development.
- To apply classroom concepts of frontend development, REST APIs and databases to a real project.
- To understand the workflow of designing and building a responsive user interface.
- To learn how frontend, backend, database and cloud storage components work together.
- To develop discipline in testing, debugging and refining a web application.
- To build confidence in independently completing a project within a fixed timeline.

1.3 Scope of the Report

This report documents the complete internship experience at Codec Technologies Pvt. Ltd., covering the organization profile, week-wise progression, detailed project work, technologies used, testing, learning outcomes, challenges and future enhancement. The project described is the MyDocs prototype developed during the internship and does not claim production-grade deployment, large-scale load handling or third-party security certification.

CHAPTER 2: ORGANIZATION PROFILE

2.1 About Codec Technologies Pvt. Ltd.

Codec Technologies Pvt. Ltd. is the organization at which I completed my internship. As per the official offer letter issued to me, Codec Technologies is described as a platform dedicated to empowering learners and connecting talent with meaningful career opportunities, offering IT and business consultancy services. The internship programme is affiliated with the National Internship Portal under the Ministry of Education and is recognized as an AICTE & ICAC approved internship programme, as reflected on the Internship Certificate.

The internship offered structured, project-based learning in the Web Development domain, allowing interns to work on assigned projects and training tasks under the guidance of designated project heads, with performance reviewed on a weekly basis.

2.2 Internship Program Details

The table below summarizes the official details of my internship as confirmed by the Internship Offer Letter and the Internship Completion Certificate.

Particular	Details
Organization	Codec Technologies Pvt. Ltd.
Designation	Web Developer Intern
Duration	18 June 2026 - 18 July 2026
Duration Length	1 Month
Mode	Hybrid
Location	Pan India
Reporting To	Assigned Project Head(s)
Domain	Web Development / Full-Stack Web Development

CHAPTER 3: WEEKLY WORK DONE

The internship progressed in a structured, week-wise manner, moving from orientation and project planning to frontend development, backend and database implementation, cloud integration, collaboration features, testing and final documentation.

3.1 Week 1: Orientation & Foundation

18 June 2026 - 24 June 2026

Main Focus: Understanding the internship workflow and establishing the technical foundation for the MyDocs project.

- Attended internship orientation and understood the expected deliverables.
- Set up the development environment using Visual Studio Code, Node.js and Git.
- Revised HTML5, CSS3 and JavaScript fundamentals, along with React and Next.js basics.
- Studied how frontend, backend, database and cloud storage work together in a full-stack application.
- Planned the structure, modules and feature set of the MyDocs project.

Skills gained

Familiarity with the development workflow and a clear technical plan for the MyDocs project.

3.2 Week 2: Frontend Development

25 June 2026 - 1 July 2026

Main Focus: Building the pages and visual design of the MyDocs interface using Next.js.

- Created the Next.js project structure with home, login, dashboard and document detail pages.
- Built the login and registration interface with API integration.
- Developed the dashboard with document listing, upload form and search bar.
- Implemented the dark-themed visual design and responsive layout of MyDocs.
- Connected the frontend to backend APIs using the Fetch API and token-based authentication.

CHAPTER 3: WEEKLY WORK DONE

3.3 Week 3: Backend & Database

02 July 2026 - 08 July 2026

Main Focus: Building the server-side logic, database models and secure cloud file storage for MyDocs.

- Set up the Express server with REST API routes for authentication and documents.
- Designed the PostgreSQL database schema using Prisma for users, documents, versions, permissions, tags, comments and audit logs.
- Implemented registration and login with password hashing and JWT tokens.
- Integrated AWS S3 for cloud file storage with AES-256 encryption before upload.
- Added document versioning, role-based sharing, tagging and audit logging.
- Implemented real-time presence, comments and edit locks using Socket.io.

3.4 Week 4: Integration, Testing & Finalization

09 July 2026 - 18 July 2026

Main Focus: Connecting the application layers, testing the major features and preparing final documentation.

- Integrated the Next.js frontend with the backend APIs end to end.
- Conducted manual testing of upload, download, versioning, sharing, search and access permissions.
- Tested real-time collaboration features with multiple logged-in sessions.
- Debugged authentication, encryption and file-handling issues found during testing.
- Refined the UI with improved buttons, icons, spacing and styling.
- Prepared project documentation, internship report material and presentation content.

Overall progression

Orientation -> Frontend -> Backend & Database -> Cloud Storage -> Collaboration -> Testing -> Documentation.

CHAPTER 4: PROJECT WORK

4.1 Project Title

Development of MyDocs: A Secure Cloud-Based Document Management System

4.2 Project Introduction

MyDocs is a full-stack, cloud-based document management system developed during the internship using Next.js, Node.js with Express, PostgreSQL and AWS S3. It is designed as a secure, organized platform where users can store documents, share them with others under controlled permissions, and collaborate in real time.

The system combines a modern Next.js interface with a REST API backend, a relational PostgreSQL database managed through Prisma, encrypted cloud file storage on AWS S3, and live collaboration features implemented using Socket.io.

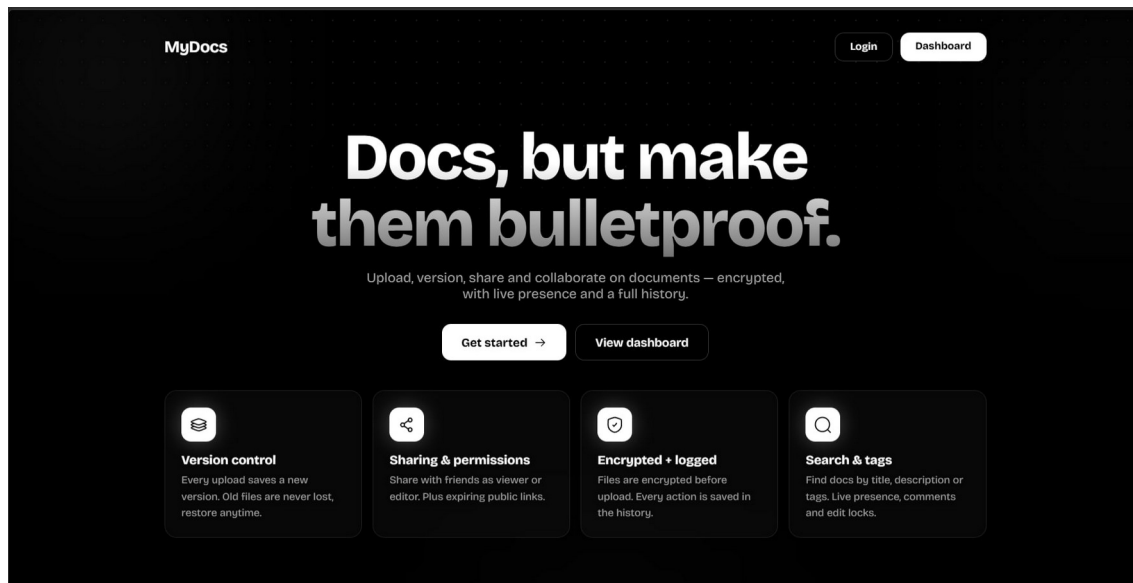


Figure 1: MyDocs Landing Page

CHAPTER 4: PROJECT WORK

4.3 Problem Statement

Many individuals and small teams manage documents through scattered files, email attachments and manual sharing. This can lead to lost or overwritten versions, uncontrolled sharing, limited visibility into who accessed or modified a file, weak protection for sensitive documents, and difficulty locating files later.

MyDocs addresses these issues through a centralized document vault with automatic version control, role-based sharing, expiring public share links, encrypted cloud storage, audit logging, real-time collaboration features, and advanced search with tagging. Together, these capabilities aim to improve the security, organization and usability of document handling.

4.4 Objectives of the Project

- Design a modern, organized document storage and sharing interface.
- Implement secure user registration and login with token-based authentication.
- Build automatic version control so every file update preserves document history.
- Implement role-based access control for controlled document sharing.
- Protect stored files through encryption and maintain an audit trail of important actions.
- Enable real-time collaboration through presence, comments and edit locks.
- Apply Next.js, Node.js, PostgreSQL and AWS S3 concepts in a practical project.

CHAPTER 4: PROJECT WORK

4.5 Scope of the Project

Implemented in the current version:

- User registration and login with hashed passwords and JWT authentication.
- Document upload with AES-256 encryption and S3 cloud storage.
- Automatic version control with per-version download.
- Role-based sharing using Owner, Editor and Viewer permissions.
- Expiring public share links.
- Real-time presence, comments and edit locks.
- Tagging and advanced search.
- Audit logging of major document and account actions.
- Soft delete with restore.

Not implemented in the current version:

- In-browser file preview for PDF, image and video files.
- Folder-based organization of documents.
- Email notifications for shares and updates.
- Production-grade deployment with load balancing.
- Third-party security audit or certification.

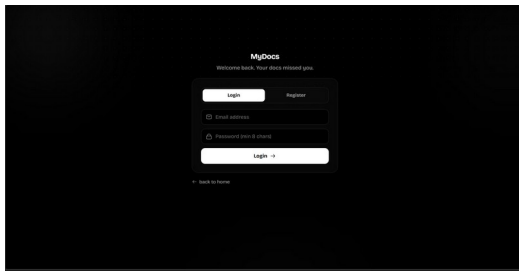


Figure 2: Login Interface

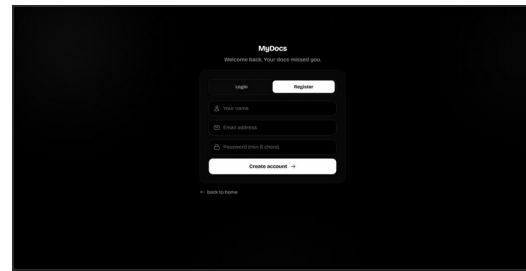


Figure 3: Registration Interface

CHAPTER 4: PROJECT WORK

4.6 Features Implemented

The MyDocs system implements a set of features focused on secure storage, controlled sharing and collaborative document handling. The table summarizes the principal features.

Feature	Description
User Registration	Captures name, email and password with validation and duplicate-email checking.
User Login	Verifies credentials and issues a JWT token for authenticated requests.
Document Upload	Accepts title, description and tags; encrypts the file before S3 upload and creates version 1.
Version Control	Every update creates a new numbered version with a changelog; previous versions can be downloaded.
Role-Based Sharing	Owner can share a document with another registered user as Editor or Viewer.
Expiring Share Links	Provides time-limited public links for controlled external sharing.
Real-Time Collaboration	Shows active viewers and supports collaborative interaction around a document.
Comments & Edit Locks	Authorized users can comment and lock a document while editing.
Search & Tagging	Documents can be tagged and located using search functionality.
Audit Logging	Records important actions to provide a trace of document activity.

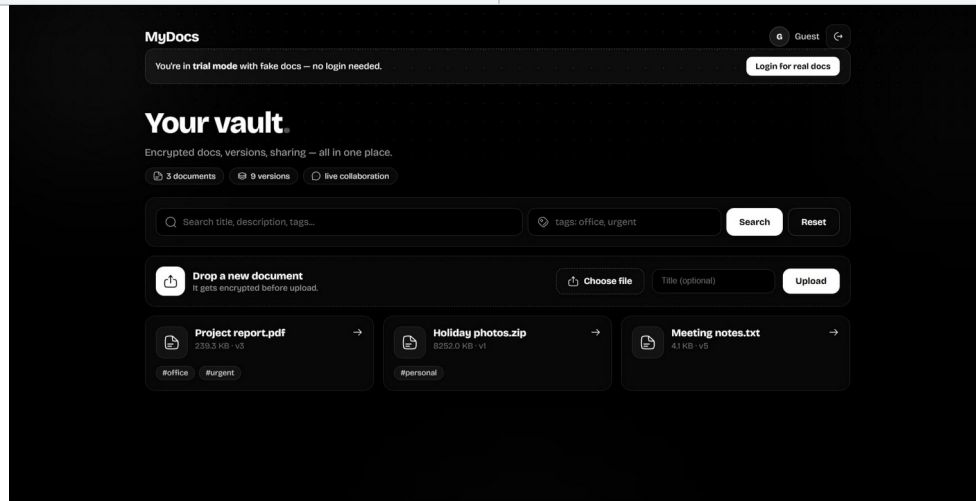


Figure 4: MyDocs Dashboard / Document Vault

CHAPTER 4: PROJECT WORK

4.7 Functional Requirements

- Authentication: users must register and log in securely before accessing protected documents.
- Document Management: authorized users must upload, download, update and manage versions.
- Access Control: owners must assign Editor or Viewer permissions.
- Sharing: controlled sharing must support permissions and expiring links.
- Encryption: uploaded document content must be encrypted before cloud storage.
- Collaboration: authorized users must see active viewers, add comments and use edit locks.
- Search and Audit: users must search by metadata and review important activity.

4.8 System Design / Architecture

MyDocs follows a layered full-stack architecture. The Next.js frontend provides the user interface, the Node.js/Express backend exposes REST APIs, PostgreSQL stores structured application data through Prisma, AWS S3 stores encrypted document objects, and Socket.io supports real-time collaboration events.

MyDocs System Architecture

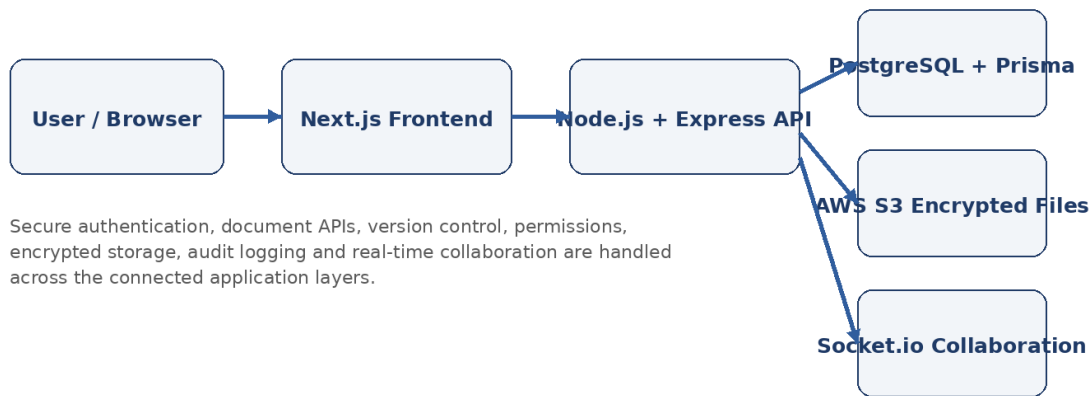
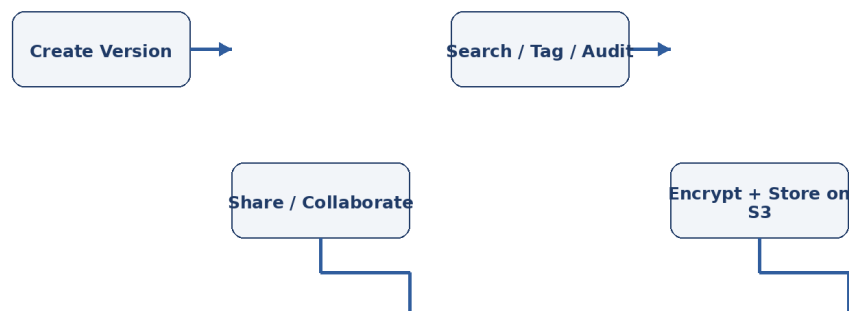


Figure 5: MyDocs System Architecture

4.9 Project Workflow

MyDocs Document Workflow



A document moves from authenticated access through secure storage and versioning, then into controlled sharing, collaboration, search and audit tracking.

Figure 6: MyDocs Document Workflow

CHAPTER 4: PROJECT WORK

4.10 Implementation Details

Next.js: Used for the responsive frontend, page routing, dashboard views, authentication screens, document views and user interactions.

Node.js with Express: Used to build the REST API layer responsible for authentication, document operations, sharing, permissions, search and other server-side logic.

PostgreSQL with Prisma: PostgreSQL stores relational data such as users, documents, versions, permissions, tags, comments and audit logs. Prisma provides the ORM layer for structured database access.

AWS S3 and Encryption: Uploaded files are encrypted before being stored as cloud objects. AWS S3 provides the document storage layer while the application manages secure upload and retrieval operations.

Socket.io: Used for real-time presence indicators, comments and edit-lock events so authorized users can receive collaboration updates without repeatedly refreshing the page.

4.11 Testing

Case	Condition	Expected Result	Status
TC-01	Invalid login credentials	Authentication request rejected and an appropriate error is shown.	Pass
TC-02	Upload valid document	File accepted, encrypted and stored; version 1 created.	Pass
TC-03	Invalid file input	Upload rejected/handled without corrupting stored data.	Pass
TC-04	Update existing document	New numbered version created; previous history retained.	Pass
TC-05	Viewer attempts restricted edit	Viewer cannot perform unauthorized operations.	Pass
TC-06	Editor accesses shared document	Editor accesses according to assigned permission.	Pass
TC-07	Expired share link	Expired link no longer provides access.	Pass
TC-08	Concurrent document viewing	Presence information updates for active users.	Pass
TC-09	Add comment	Comment becomes visible to authorized viewers.	Pass
TC-10	Search by title/tag	Matching documents returned from the search operation.	Pass
TC-11	Soft delete and restore	Deleted document leaves active view and can be restored.	Pass
TC-12	Audit log check	Important actions appear in activity history.	Pass

CHAPTER 4: PROJECT WORK

4.12 Results / Output

The completed MyDocs prototype provides a centralized document-management workflow covering authentication, secure upload, version control, controlled sharing, cloud storage, collaboration, search and audit tracking. The implementation demonstrates how a modern full-stack application can combine a responsive frontend with REST APIs, relational data management and cloud object storage.

The project output demonstrates the intended end-to-end flow from user authentication to document storage and collaboration. Testing confirmed the major functional paths used during development, while the remaining limitations are identified as future enhancements rather than completed production capabilities.

MyDocs Core Features



Figure 7: MyDocs Core Features

Observed outcome

The prototype consolidates document handling into a single workflow while keeping access, version history and activity visibility explicit.

CHAPTER 5: TOOLS & TECHNOLOGIES USED

The project combines the following technologies into a single full-stack workflow. Each component was selected according to the role it performs in the application.

Category	Tool / Technology
Frontend Framework	Next.js - application interface, routing and React-based components
Backend Runtime / Framework	Node.js with Express - REST API and server-side application logic
Database	PostgreSQL - relational storage for users, documents, permissions, versions, tags, comments and audit records
ORM	Prisma - database schema and structured data access
Cloud Storage	AWS S3 - cloud object storage for encrypted document files
Security	Bcrypt, JWT and Node.js crypto - password hashing, authentication and file encryption support
Real-Time Communication	Socket.io - presence, comments and edit-lock events
Development Environment	Visual Studio Code and Git - coding, project management and version tracking
Technology flow	
Next.js UI -> Express API -> PostgreSQL / Prisma + AWS S3 -> Socket.io collaboration events.	

CHAPTER 6: LEARNING OUTCOMES

6.1 Technical Skills

- Building responsive interfaces using Next.js and React concepts.
- Connecting frontend applications to REST APIs using authenticated requests.
- Designing relational database structures using PostgreSQL and Prisma.
- Implementing user authentication using password hashing and JWT.
- Working with cloud object storage through AWS S3.
- Understanding file encryption and secure document handling workflows.
- Implementing version control, role-based access permissions and audit logging.
- Using Socket.io concepts for real-time presence, comments and edit locks.
- Testing, debugging and integrating multiple application layers.

6.2 Professional / Soft Skills

- Time management while completing project milestones within a fixed one-month duration.
- Understanding and interpreting project requirements independently.
- Problem-solving while resolving integration, authentication and file-handling issues.
- Planning tasks in a structured, week-wise manner.
- Preparing clear technical documentation.
- Presenting completed work in a structured, professional format.

CHAPTER 7: CHALLENGES FACED

- Designing a clean document-management interface while keeping dashboard information easy to navigate.
- Maintaining consistent communication between the Next.js frontend and the Express REST API.
- Designing relational database models that correctly represent users, documents, versions, permissions, tags, comments and audit records.
- Handling authentication and authorization consistently across protected API routes.
- Managing encrypted file uploads and cloud storage without breaking retrieval and versioning behaviour.
- Implementing role-based permissions so Owner, Editor and Viewer actions remain appropriately separated.
- Keeping real-time presence, comments and edit-lock states synchronized across multiple sessions.
- Debugging issues that appeared only when frontend, backend, database and storage components were used together.
- Testing sharing, versioning, search and soft-delete workflows across different user and document states.

Key takeaway

The main challenge was not any single technology, but making several application layers behave as one coherent system.

CHAPTER 8: FUTURE SCOPE

The current MyDocs prototype establishes the core document-management workflow, but several enhancements can make the system more complete and suitable for broader real-world use.

- In-browser preview for PDF, image and video documents.
- Folder-based organization and hierarchical document collections.
- Email notifications for document sharing, comments, updates and access events.
- Advanced search filters and richer document metadata.
- Granular permission policies and organization-level access management.
- Production-grade deployment with load balancing, monitoring, backups and automated scaling.
- Additional security controls such as multi-factor authentication and stronger key-management practices.
- Comprehensive automated testing and third-party security auditing.
- Subscription or payment management if the system is later offered as a commercial service.

Future direction

The natural next step is to move from a functional prototype to a production-ready collaborative document platform with stronger operational, security and usability layers.

CHAPTER 9: CONCLUSION

The one-month internship as a Web Developer Intern at Codec Technologies Pvt. Ltd., undertaken from 18 June 2026 to 18 July 2026, provided valuable hands-on experience in full-stack web development. Working in the Web Development / Full-Stack Web Development domain allowed me to move beyond theoretical classroom learning and apply frontend, backend, database and cloud concepts to a complete project.

Through the development of MyDocs, a secure cloud-based document management system, I gained practical experience in building a responsive Next.js interface, designing REST APIs with Node.js and Express, modelling relational data using PostgreSQL and Prisma, integrating AWS S3 for cloud storage, and working with authentication, encryption, access permissions and real-time collaboration.

The process of integrating, testing and debugging the different application layers strengthened my understanding of how modern web systems are designed and maintained. It also improved my problem-solving, time-management, documentation and presentation skills.

Overall, this internship helped bridge the gap between academic learning and practical software development. The MyDocs project provided a strong foundation for further work in full-stack engineering, cloud applications, secure document systems and collaborative web platforms.

Final outcome

A structured prototype demonstrating document management, secure storage, controlled sharing and collaborative workflows within one integrated system.

REFERENCES

Primary Sources

- Codec Technologies Pvt. Ltd. - Internship Offer Letter, issued 18 June 2026.
- Codec Technologies Pvt. Ltd. - Internship Completion Certificate, National Internship Portal, Ministry of Education.

Technical References

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- Socket.io Documentation - socket.io/docs

Report note

The project descriptions, figures and test cases in this report are based on the project draft and supporting visuals supplied for the internship report.